

ECEN 629 Applied Convex Optimization

Chapter 11 Inequality constrained minimization

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Inequality constrained minimization

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b \end{array} \quad (1)$$

- f_i convex, twice continuously differentiable
- $A \in \mathbf{R}^{p \times n}$ with $\text{rank } A = p$
- we assume p^* is finite and attained
- we assume problem is strictly feasible: there exists $\tilde{x}$ with

$$\tilde{x} \in \text{dom } f_0, \quad f_i(\tilde{x}) < 0, \quad i = 1, \dots, m, \quad A\tilde{x} = b$$

hence, strong duality holds and dual optimum is attained

Examples

- LP, QP, QCQP, GP
- entropy maximization with linear inequality constraints

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^n x_i \log x_i \\ \text{subject to} & Fx \preceq g \\ & Ax = b\end{array}$$

with $\text{dom } f_0 = \mathbf{R}_{++}^n$

- differentiability may require reformulating the problem, *e.g.*, piecewise-linear minimization or ℓ_∞ -norm approximation via LP
- SDPs and SOCPs are better handled as problems with generalized inequalities (see later)

Logarithmic barrier

reformulation of (1) via indicator function:

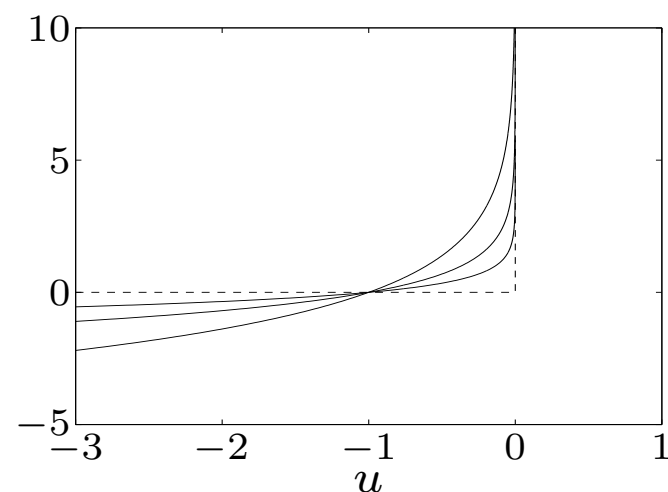
$$\begin{array}{ll} \text{minimize} & f_0(x) + \sum_{i=1}^m I_-(f_i(x)) \\ \text{subject to} & Ax = b \end{array}$$

where $I_-(u) = 0$ if $u \leq 0$, $I_-(u) = \infty$ otherwise (indicator function of $\mathbf{R}_-$)

approximation via logarithmic barrier

$$\begin{array}{ll} \text{minimize} & f_0(x) - (1/t) \sum_{i=1}^m \log(-f_i(x)) \\ \text{subject to} & Ax = b \end{array}$$

- an equality constrained problem
- for $t > 0$, $-(1/t) \log(-u)$ is a smooth approximation of I_-
- approximation improves as $t \rightarrow \infty$



logarithmic barrier function

$$\phi(x) = -\sum_{i=1}^m \log(-f_i(x)), \quad \mathbf{dom} \phi = \{x \mid f_1(x) < 0, \dots, f_m(x) < 0\}$$

- convex (follows from composition rules)
- twice continuously differentiable, with derivatives

$$\nabla \phi(x) = \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x)$$

$$\nabla^2 \phi(x) = \sum_{i=1}^m \frac{1}{f_i(x)^2} \nabla f_i(x) \nabla f_i(x)^T + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla^2 f_i(x)$$

Central path

- for $t > 0$, define $x^*(t)$ as the solution of

$$\begin{array}{ll}\text{minimize} & t f_0(x) + \phi(x) \\ \text{subject to} & Ax = b\end{array}$$

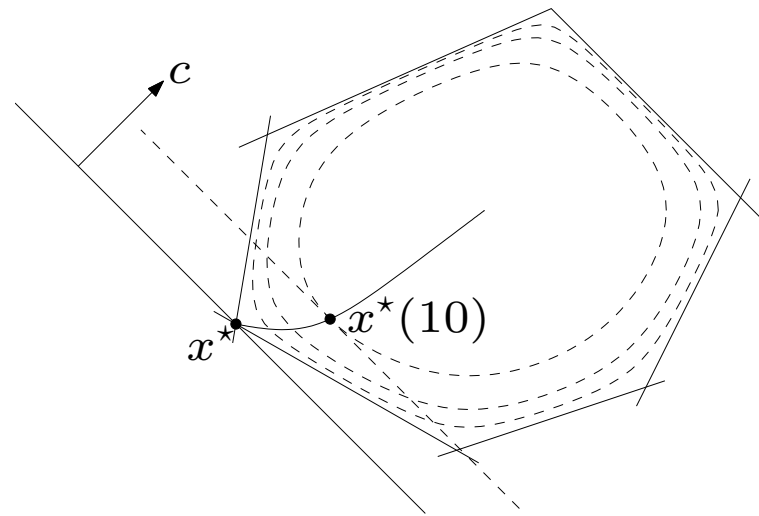
(for now, assume $x^*(t)$ exists and is unique for each $t > 0$)

- central path is $\{x^*(t) \mid t > 0\}$

example: central path for an LP

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, 6\end{array}$$

hyperplane $c^T x = c^T x^*(t)$ is tangent to level curve of ϕ through $x^*(t)$



Dual points on central path

$x = x^*(t)$ if there exists a w such that

$$t\nabla f_0(x) + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x) + A^T w = 0, \quad Ax = b$$

- therefore, $x^*(t)$ minimizes the Lagrangian

$$L(x, \lambda^*(t), \nu^*(t)) = f_0(x) + \sum_{i=1}^m \lambda_i^*(t) f_i(x) + \nu^*(t)^T (Ax - b)$$

where we define $\lambda_i^*(t) = 1/(-tf_i(x^*(t)))$ and $\nu^*(t) = w/t$

- this confirms the intuitive idea that $f_0(x^*(t)) \rightarrow p^*$ if $t \rightarrow \infty$:

$$\begin{aligned} p^* &\geq g(\lambda^*(t), \nu^*(t)) \\ &= L(x^*(t), \lambda^*(t), \nu^*(t)) \\ &= f_0(x^*(t)) - m/t \end{aligned}$$

Interpretation via KKT conditions

$x = x^*(t)$, $\lambda = \lambda^*(t)$, $\nu = \nu^*(t)$ satisfy

1. primal constraints: $f_i(x) \leq 0$, $i = 1, \dots, m$, $Ax = b$
2. dual constraints: $\lambda \succeq 0$
3. approximate complementary slackness: $-\lambda_i f_i(x) = 1/t$, $i = 1, \dots, m$
4. gradient of Lagrangian with respect to x vanishes:

$$\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + A^T \nu = 0$$

difference with KKT is that condition 3 replaces $\lambda_i f_i(x) = 0$

Force field interpretation

centering problem (for problem with no equality constraints)

$$\text{minimize } tf_0(x) - \sum_{i=1}^m \log(-f_i(x))$$

force field interpretation

- $tf_0(x)$ is potential of force field $F_0(x) = -t\nabla f_0(x)$
- $-\log(-f_i(x))$ is potential of force field $F_i(x) = (1/f_i(x))\nabla f_i(x)$

the forces balance at $x^*(t)$:

$$F_0(x^*(t)) + \sum_{i=1}^m F_i(x^*(t)) = 0$$

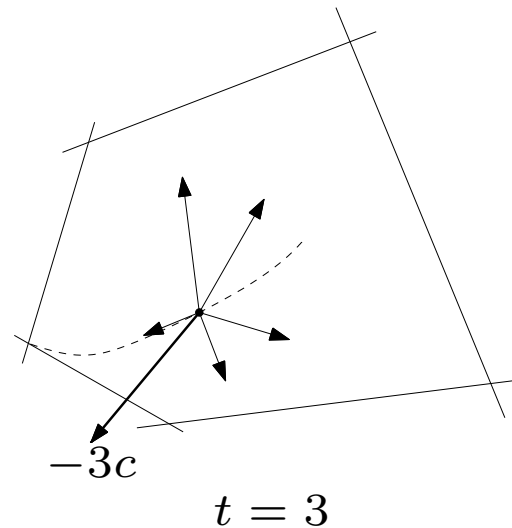
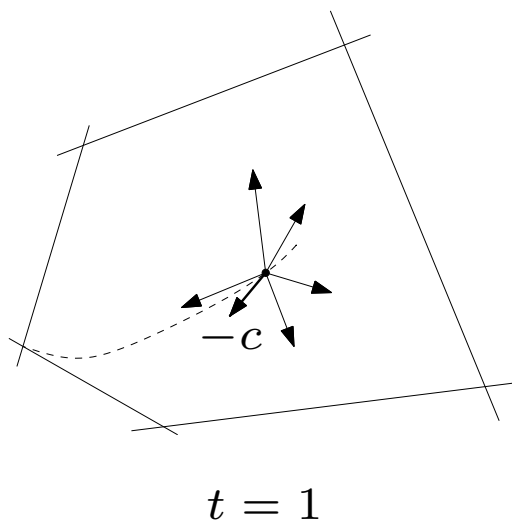
example

$$\begin{array}{ll}\text{minimize} & c^T x \\ \text{subject to} & a_i^T x \leq b_i, \quad i = 1, \dots, m\end{array}$$

- objective force field is constant: $F_0(x) = -tc$
- constraint force field decays as inverse distance to constraint hyperplane:

$$F_i(x) = \frac{-a_i}{b_i - a_i^T x}, \quad \|F_i(x)\|_2 = \frac{1}{\mathbf{dist}(x, \mathcal{H}_i)}$$

where $\mathcal{H}_i = \{x \mid a_i^T x = b_i\}$



Barrier method

given strictly feasible x , $t := t^{(0)} > 0$, $\mu > 1$, tolerance $\epsilon > 0$.

repeat

1. *Centering step.* Compute $x^*(t)$ by minimizing $tf_0 + \phi$, subject to $Ax = b$.
 2. *Update.* $x := x^*(t)$.
 3. *Stopping criterion.* **quit** if $m/t < \epsilon$.
 4. *Increase t .* $t := \mu t$.
-

- terminates with $f_0(x) - p^* \leq \epsilon$ (stopping criterion follows from $f_0(x^*(t)) - p^* \leq m/t$)
- centering usually done using Newton's method, starting at current x
- choice of μ involves a trade-off: large μ means fewer outer iterations, more inner (Newton) iterations; typical values: $\mu = 10$ – 20
- several heuristics for choice of $t^{(0)}$

Convergence analysis

number of outer (centering) iterations: exactly

$$\left\lceil \frac{\log(m/(\epsilon t^{(0)}))}{\log \mu} \right\rceil$$

plus the initial centering step (to compute $x^*(t^{(0)})$)

centering problem

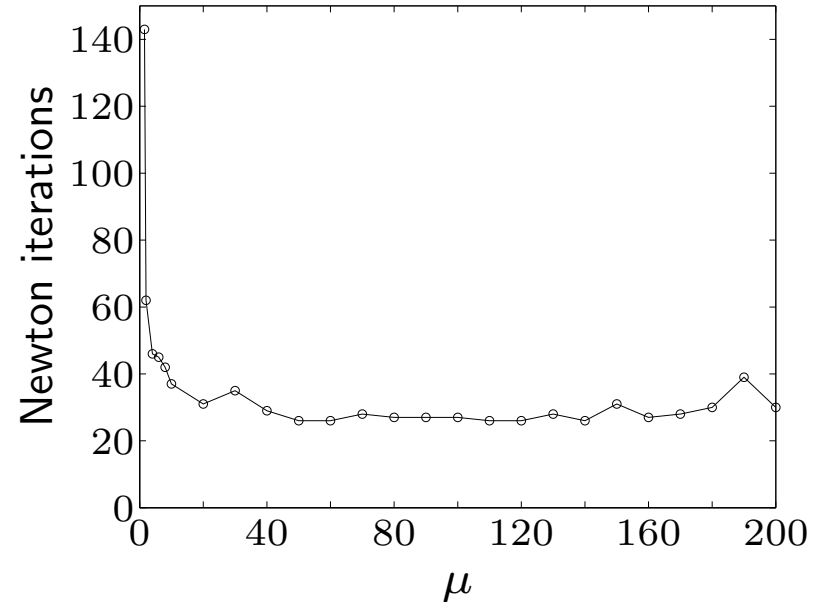
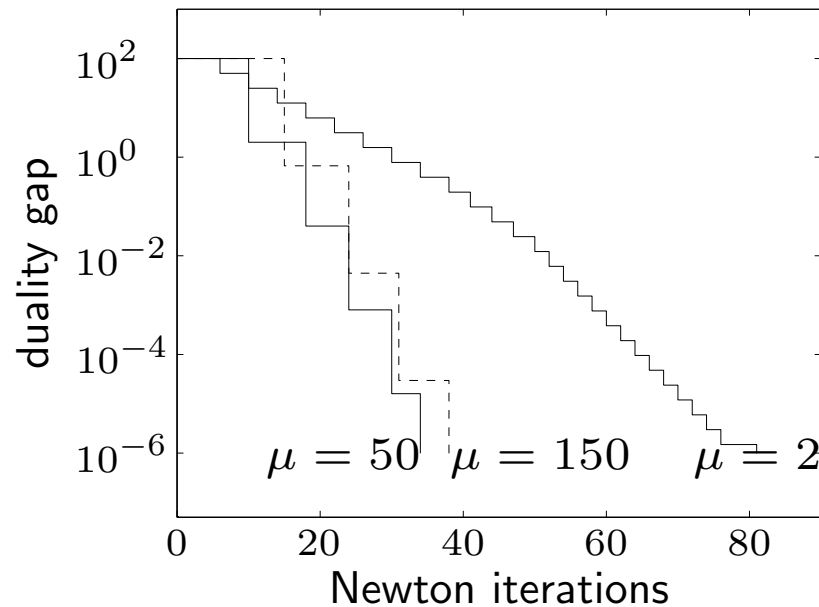
$$\text{minimize } tf_0(x) + \phi(x)$$

see convergence analysis of Newton's method

- $tf_0 + \phi$ must have closed sublevel sets for $t \geq t^{(0)}$
- classical analysis requires strong convexity, Lipschitz condition
- analysis via self-concordance requires self-concordance of $tf_0 + \phi$

Examples

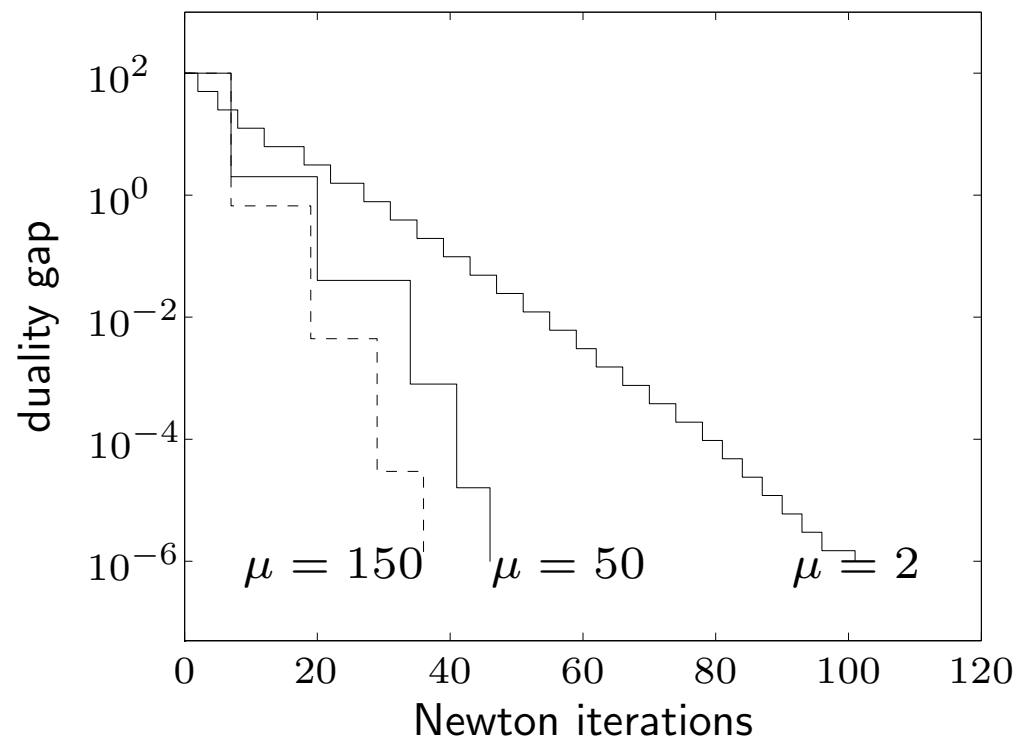
inequality form LP ($m = 100$ inequalities, $n = 50$ variables)



- starts with x on central path ($t^{(0)} = 1$, duality gap 100)
- terminates when $t = 10^8$ (gap 10^{-6})
- centering uses Newton's method with backtracking
- total number of Newton iterations not very sensitive for $\mu \geq 10$

geometric program ($m = 100$ inequalities and $n = 50$ variables)

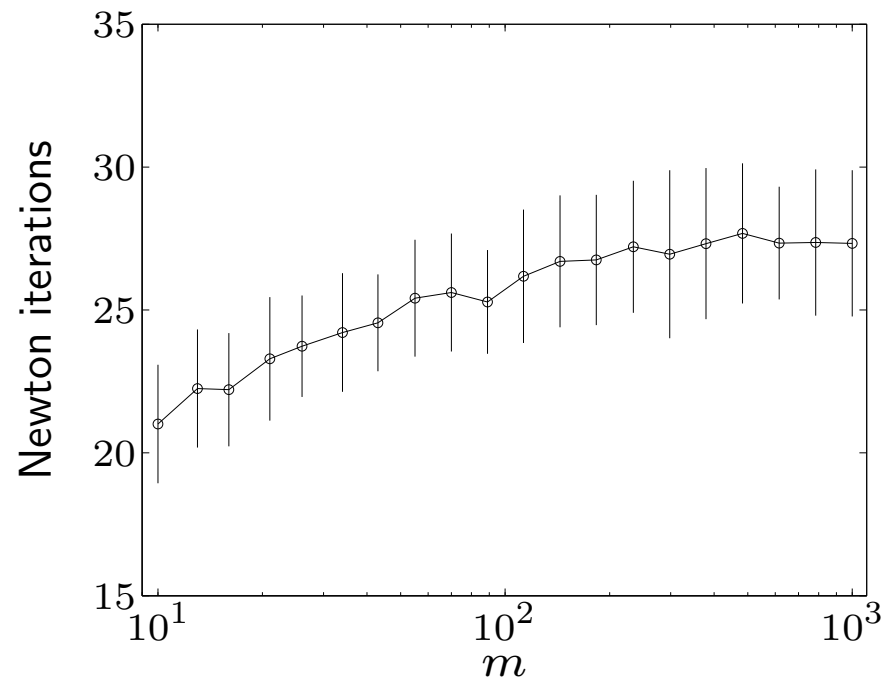
$$\begin{array}{ll} \text{minimize} & \log \left(\sum_{k=1}^5 \exp(a_{0k}^T x + b_{0k}) \right) \\ \text{subject to} & \log \left(\sum_{k=1}^5 \exp(a_{ik}^T x + b_{ik}) \right) \leq 0, \quad i = 1, \dots, m \end{array}$$



family of standard LPs ($A \in \mathbf{R}^{m \times 2m}$)

$$\begin{array}{ll} \text{minimize} & c^T x \\ \text{subject to} & Ax = b, \quad x \succeq 0 \end{array}$$

$m = 10, \dots, 1000$; for each m , solve 100 randomly generated instances



number of iterations grows very slowly as m ranges over a 100 : 1 ratio

Exercise 1: The Newton Step

Find the Newton step for the following optimization problem in the log-barrier algorithm when $t = 10$

$$\begin{aligned} &\text{minimize: } f(x) = \log(e^{x_1 - 3x_2} + e^{2x_1 + x_2} + 2) \\ &\text{subject to: } x_1^2 - 3x_1x_2 + 8x_2^2 \leq 5, \end{aligned}$$

where the current point is $x^{(k)} = (1, 0.5)^T$.

Feasibility and phase I methods

feasibility problem: find x such that

$$f_i(x) \leq 0, \quad i = 1, \dots, m, \quad Ax = b \quad (2)$$

phase I: computes strictly feasible starting point for barrier method

basic phase I method

$$\begin{array}{ll} \text{minimize (over } x, s) & s \\ \text{subject to} & f_i(x) \leq s, \quad i = 1, \dots, m \\ & Ax = b \end{array} \quad (3)$$

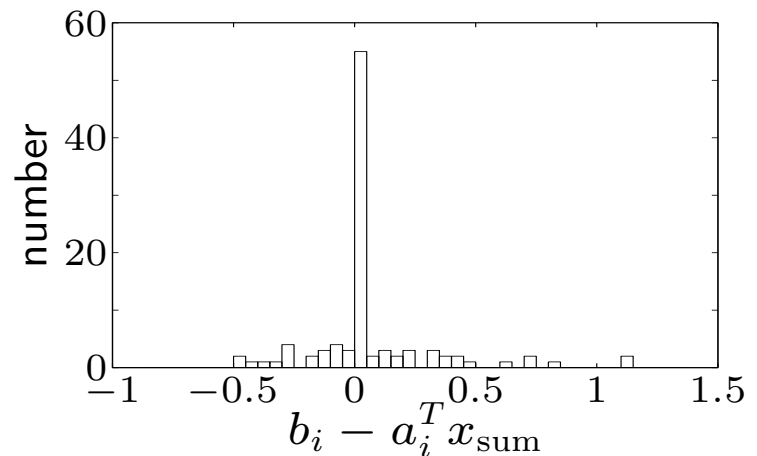
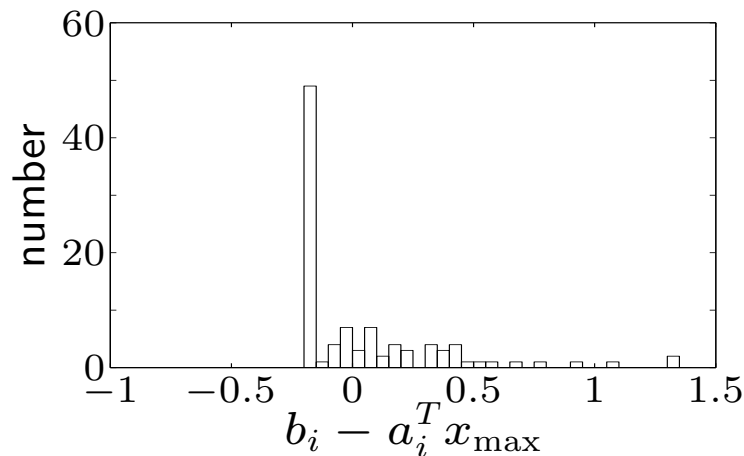
- if x, s feasible, with $s < 0$, then x is strictly feasible for (2)
- if optimal value $\bar{p}^*$ of (3) is positive, then problem (2) is infeasible
- if $\bar{p}^* = 0$ and attained, then problem (2) is feasible (but not strictly);
if $\bar{p}^* = 0$ and not attained, then problem (2) is infeasible

sum of infeasibilities phase I method

$$\begin{array}{ll}\text{minimize} & \mathbf{1}^T s \\ \text{subject to} & s \succeq 0, \quad f_i(x) \leq s_i, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

for infeasible problems, produces a solution that satisfies many more inequalities than basic phase I method

example (infeasible set of 100 linear inequalities in 50 variables)

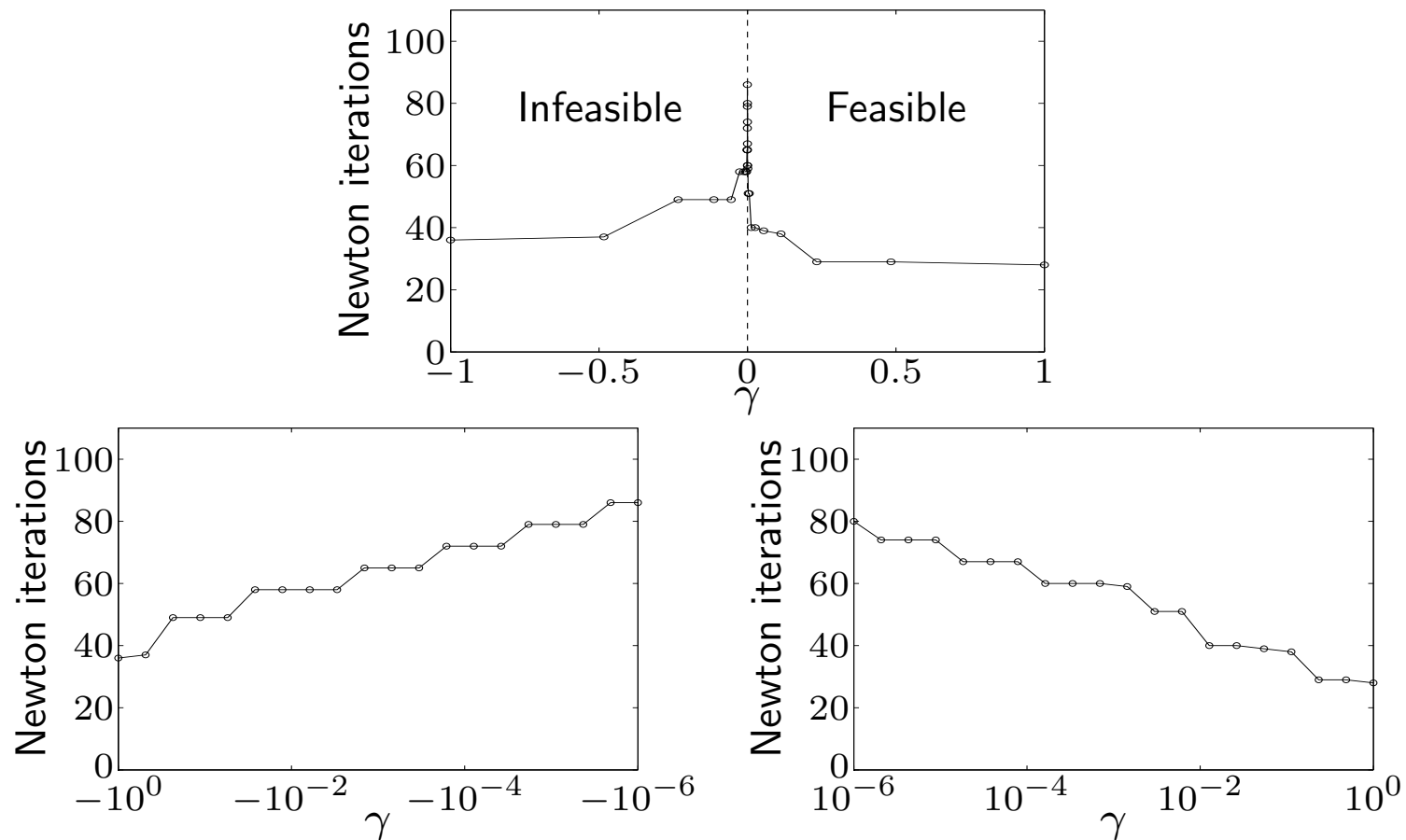


left: basic phase I solution; satisfies 39 inequalities

right: sum of infeasibilities phase I solution; satisfies 79 inequalities

example: family of linear inequalities $Ax \preceq b + \gamma \Delta b$

- data chosen to be strictly feasible for $\gamma > 0$, infeasible for $\gamma \leq 0$
- use basic phase I, terminate when $s < 0$ or dual objective is positive



number of iterations roughly proportional to $\log(1/|\gamma|)$

Primal-dual interior-point method

Rewrite the modified KKT conditions

$$r_t(x, \lambda, \nu) = \begin{bmatrix} \nabla f_0(x) + Df(x)^T \lambda + A^T \nu \\ -\mathbf{diag}(\lambda)f(x) - (1/t)\mathbf{1} \\ Ax - b \end{bmatrix}$$

- First block: dual residual;
- Second block: centrality residual;
- Last block: primal residual.

Newton step from modified KKT conditions

Through Newton step: $r_t(y + \Delta y) \approx r_t(y) + Dr_t(y)\Delta y = 0 \Rightarrow$

$$\begin{bmatrix} \nabla^2 f_0(x) + \sum_{i=1}^m \lambda_i \nabla^2 f_i(x) & Df(x)^T & A^T \\ -\mathbf{diag}(\lambda) Df(x) & -\mathbf{diag}(f(x)) & 0 \\ A x & 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta \lambda \\ \Delta \nu \end{bmatrix} = - \begin{bmatrix} r_{dual} \\ r_{cent} \\ r_{pri} \end{bmatrix}$$

$\Rightarrow$ the primal-dual search direction $\Delta y_{pd} = (\Delta x, \Delta \lambda, \Delta \nu)$.

Comparison with barrier method

The Newton steps of the two methods are very similar

- Barrier method:

$$\begin{bmatrix} H_{bar} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{pd} \\ \Delta \nu_{pd} \end{bmatrix} = - \begin{bmatrix} \nabla f_0(x) - (1/t) \sum_{i=1}^m \frac{\nabla f_i(x)}{f_i(x)} + A^T \nu \\ r_{pri} \end{bmatrix}$$

where

$$H_{bar} = \nabla^2 f_0(x) + \sum_{i=1}^m \lambda_i \nabla^2 f_i(x) - \sum_{i=1}^m \frac{\lambda_i}{f_i(x)} \nabla f_i(x) \nabla f_i(x)^T.$$

- Primal-dual interior:

$$\begin{bmatrix} H_{pd} & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{pd} \\ \Delta \nu_{pd} \end{bmatrix} = - \begin{bmatrix} \nabla f_0(x) - (1/t) \sum_{i=1}^m \frac{\nabla f_i(x)}{f_i(x)} + A^T \nu \\ r_{pri} \end{bmatrix}$$

where

$$H_{pd} = \nabla^2 f_0(x) + \sum_{i=1}^m \frac{1}{-t f_i(x)} \nabla^2 f_i(x) - \sum_{i=1}^m \frac{1}{t f_i^2(x)} \nabla f_i(x) \nabla f_i(x)^T.$$

The surrogate duality gap

The intermediate points $x^{(k)}$, $\lambda^{(k)}$, $\nu^{(k)}$ are not necessarily feasible.

- Define: $\hat{\eta}(x, \lambda) = -f(x)^T \lambda$, the surrogate duality gap;
- Equal to the duality gap when the points are feasible.

Primal-dual interior-point method

Given x , $f_i(x) < 0$ for $i = 1, 2, \dots, m$, $\lambda \succ 0$ and $\nu > 0$, $\epsilon_{fea} > 0$, and $\epsilon > 0$.

Repeat

1. Determine t . Set $t = \mu m / \hat{\eta}$.
2. Compute the primal-dual search direction Δy_{pd} .
3. Line search and update.

Until $\|r_{pri}\|_2 \leq \epsilon_{fea}$, $\|r_{dual}\|_2 \leq \epsilon_{fea}$, and $\hat{\eta} \leq \epsilon$.

Can be viewed as infeasible start barrier method.